Question Paper

Exam Date & Time: 27-Dec-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.Tech. (ELECTRONICS & COMMUNICATION ENGINEERING) EXAMINATIONS - DECEMBER 2024 / JANUARY 2025
SUBJECT: ECE 3121/ECE_3121- ANALOG AND DIGITAL COMMUNICATION
(MAKEUP)

Marks: 50 **Duration: 180 mins.**

Answer all the questions.

1A) What is Single Sideband (SSB) modulation, and why is it preferred over traditional AM for communication? Explain the working methodology of Balanced modulator with neat block diagram and relevant mathematical expressions. (5)

1B) For an AM signal with a carrier frequency of 1 MHz, a modulating frequency of 10 kHz, calculate the frequencies of the sidebands and the bandwidth of the signal. How does the modulation index affect the sideband amplitudes? (3)

1C) Obtain the Fourier transform of $f(t) = e^{-2t}u(t)$, where u(t) is the unit step function. Using the time-shifting property, obtain the Fourier transform of $g(t) = e^{-2(t-1)}u(t-1)$. (2)

2A) Consider three signals $s_1(t)$, $s_2(t)$ and $s_3(t)$ defined over the interval $0 \le t \le T$. Determine the orthonormal basis functions and express the signals in terms of basis functions. Also plot the resulting orthonormal basis functions. (5)

$$s_1(t) = \begin{cases} 1, & for\ 0 \le t \le {}^{2T}/_{3} \\ 0, & Otherwise \end{cases}$$

$$s_2(t) = \begin{cases} 1, & for\ 0 \le t \le {}^{T}/_{3} \\ 0, & Otherwise \end{cases}$$

$$s_3(t) = \begin{cases} 1, & for\ {}^{T}/_{3} \le t \le {}^{2T}/_{3} \\ 0, & Otherwise \end{cases}$$

2B) Explain and derive the noise power at the output of a matched filter when the input is corrupted by AWGN with a power spectral density $N_0/2$. Assume the signal has a duration T. (3)

2C) Assume that two signals $s_1(t)$ and $s_2(t)$ are represented using two orthonormal basis functions $\Phi_1(t)$ and $\Phi_2(t)$. The coordinates of these signals are $s_1=(4,1)$ and $s_2=(-2,3)$. Determine if the signals $s_1(t)$ and $s_2(t)$ are linearly dependent. If they are dependent, express one signal as a linear combination of the other. (2)

3A) Consider the binary data sequence **1 0 0 0 1 1 0** which is applied to the input of a duo-binary system. (5)
i) Determine the output of the duo-binary coder and the corresponding receiver output, without a precoder.
ii) Determine the output of the duo-binary coder and the corresponding receiver output, with a precoder.
Assume reference bit to be 1.

3B) Determine the power spectral density of the polar NRZ code format. Assume statistically independent and equally likely message bits. (3)

3C) With a neat diagram explain the working principle of Delta modulation (DM transmitter) system. (2)

4A) Derive the expression for Nyquist criterion for distortionless baseband transmission and discuss the ideal solution for the same. (5)

4B)	Derive the expression for the average probability of symbol error for coherent binary PSK detection with all intermediate steps. Assume AWGN channel.	(3)
4C)	The binary data sequence **0 0 1 1 1 1 0 0** is applied to the input of a DPSK transmitter. Determine the differentially encoded sequence and transmitted phase. **Assume initial reference bit as 1**	(2)
5A)	Construct minimum variance Huffman code for the given source probabilities **pk=[0.1,0.11,0.19,0.21,0.39]**. Also determine the average code length of this Huffman code.	(5)
5B)	Determine the entropy, efficiency and redundancy of the Huffman code as given in Q5A.	(3)
5C)	Determine the minimum variance of this Huffman code as given in Q5A.	(2)

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